Andrew Bodling, PhD

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GPU Performance Engineer | HPC / Scientific Software Engineer

SUMMARY

- GPU-accelerated scientific software developer with 10+ years building high-throughput HPC simulation codes for DoD and industry.
- Architect and lead developer of a performance-portable, overset structured-curvilinear compressible near-body CFD solver (134M cells/s throughput) now being integrated into CREATE-AV Helios.
- Strong C++17/20, CUDA, Fortran, Python, MPI/OpenMP; experience with performance-portability frameworks (Kokkos, RAJA, OpenACC) and GPU profiling (Nsight Compute/Systems).
- Delivered measurable impact: automated end-to-end mesh/input generation cutting analyst setup time ~95%; achieved 6x speed-ups and large cost reductions on production rotorcraft/tiltrotor workflows.
- Award-winning researcher/developer (Vertical Flight Society Schroers Award) with peer-reviewed publications in rotor wake aerodynamics, aeroacoustics, and CFD methodology.

CORE SKILLS

GPU & Performance: CUDA, GPU acceleration, performance portability (Kokkos, RAJA, OpenACC), profiling (Nsight Compute/Systems)

Parallel & HPC: MPI, OpenMP, Slurm/PBS, Linux development, debugging/profiling workflows

Languages: C++17/20, Fortran, Python, Bash, MATLAB

Tooling: CMake, Git/SVN, VS Code, Pybind11

Domain: Structured-curvilinear FV/FD solvers, overset/Chimera methods, compressible flows, rotorcraft aeromechanics

PROFESSIONAL EXPERIENCE

CFD Developer - Army Computational Aeromechanics Lab (NASA Ames) | Moffett Field, CA | Jun 2019 - Present

- Led design + optimization of a single-source GPU/CPU C++ codebase (shared kernels across backends), driving end-to-end throughput to 134M cells/s (up to ~2x faster than comparable GPU solvers) and integrating into a production multi-physics framework.
- Rebuilt the Python/C++/Fortran interface that connects CREATE-AV Helios to OVERFLOW (migrated from SWIG to Pybind11) and updated Fortran glue code to call GPU routines, enabling GPU near-body CFD in Helios's overset rotorcraft workflow.
- Engineered reproducible automation pipeline that cut “time-to-first-run” (analyst setup) by 95%, improving iteration speed and reliability of GPU workflows at scale.
- Optimized and automated a HPC simulation workflow end-to-end, achieving ~6x speedup and ~83% compute-cost reduction while maintaining accuracy (~5% vs flight test).
- Led rotor-wake accuracy campaign to isolate sources of numerical error and improve prediction of rotor wake vortex systems, achieving unprecedented agreement with stereo-PIV measurements.

- Led high-fidelity CFD simulations of six-degree-of-freedom store separation for the Apache helicopter and delivered validated predictions informing design decisions (e.g., Sikorsky FARA competitive prototype).

CFD Developer - AFRL Computational Sciences Center | Dayton, OH | Jul 2018 - Jun 2019

- Integrated Mean-Flow Perturbation (MFP) analysis into FDL3DI, enabling fast embedded linear-stability studies of complex 3D flows.
- Applied MFP to a Re = 2 x 10^5, AR = 4 wingtip vortex, isolating the least-stable mode and frequency at the trailing edge and within the incipient vortex core to guide future flow-control actuator design.

Graduate Research Assistant - Computational Aerodynamics & Aeroacoustics Lab (Iowa State University) | Ames, IA | Jan 2016 - May 2019

- Extended FDL3DI to handle 90-degree corners via automated stair-step edge detection and adaptive filter order, enabling accurate LES on complex geometries.
- Re-engineered OpenMP load balancing, cutting simulation wall-clock time ~55% on a 2,000-core cluster.
- Validated Ffowcs Williams-Hawkings solver and applied it to trailing-edge noise-reduction designs; pioneered wall-resolved LES studies of bio-inspired finlet fences with validated noise reductions.

CFD Application Engineer Intern - Exa Corporation | Burlington, MA | Jul 2015 - Jan 2016

- Prepared watertight heavy-vehicle meshes (ANSA, PowerDELTA) and ran parametric PowerFLOW CFD studies across design variants, delivering data-driven drag-trade guidance to customers.
- Performed PowerFLOW simulations to optimize commercial-bus cabin airflow and built VB/Python scripts to auto-generate CFD result presentations, cutting report preparation from days to hours.

Research Assistant - Experimental Aerodynamics Lab and Computational Reliability Lab | Ames, IA | Sep 2012 - May 2015

- Performed PIV studies and implemented memory-less Latin Hypercube Sampling in the FAA SMART fracture-risk tool.

EDUCATION

PhD, Aerospace Engineering - Iowa State University | Aug 2019

Thesis: Numerical Investigations of Nearly Silent Blade Designs Inspired by the Owl

MS, Aerospace Engineering - Iowa State University | May 2017

BS, Mechanical Engineering - The University of Texas at San Antonio | Dec 2013

HONORS & SELECTED PUBLICATIONS

- Vertical Flight Society Schroers Award for Outstanding Research (GPU CFD solver development), 2024
- **Bodling, A.** et al., “Development of a Performance-Portable Compressible Flow Solver on Structured Curvilinear Grids,” *AIAA Aviation Forum*, 2024.
- **Bodling, A.** et al., “Enhancing Numerical Accuracy in the Prediction of Rotor Wake Vortex Structures,” *Physics of Fluids*, 2024.
- Full publication list: bodlinga.github.io